Comparative Risk Detection Across Banking Problems: Fraud, Money Laundering and Customer Churn

Borshon Dopho ID: 011221337 **M. A. Al Nahian** ID: 011201416 **Fahim Faysal** ID: 0112230224 **Jakia Tunnesa Rifa** ID: 0112230469

Department of Computer Science & Engineering, United International University, Dhaka, Bangladesh

***Abstract*—**Banks face several distinct risks, including fraudulent transactions, money laundering and customer attrition, yet each is usually studied in isolation with its own model and evaluation protocol. This paper compares Decision Tree, Random Forest and XGBoost classifiers on three publicly available banking datasets—PaySim (fraud, 6.36M transactions), SAML-D (money laundering, 9.50M transactions) and a Botswana bank customer churn dataset (115,640 customers)—using one shared experimental framework. The framework combines dataset-specific feature engineering, leakage control, a 70/30 train–test split, SMOTE applied only inside the training folds of a stratified five-fold cross-validation, and model selection by PR-AUC followed by F1-score. The selected models are then evaluated once on an untouched test set. Random Forest was selected for PaySim (test F1 = 99.98 %, PR-AUC = 99.99 %), and XGBoost for SAML-D (F1 = 82.38 %, PR-AUC = 93.59 %) and churn (F1 = 97.71 %, PR-AUC = 99.86 %). The SAML-D result shows a clear precision–recall trade-off (precision 74.13 %, recall 92.70 %) that overall accuracy (99.96 %) completely hides. The results indicate that no single model is guaranteed to be best across banking risks: the most suitable choice depends on the dataset, the nature of the risk, and the balance between missed cases and false alarms.

***Index Terms*—**fraud detection, anti-money laundering, customer churn, class imbalance, SMOTE, XGBoost, Random Forest, precision–recall.

I. Introduction

Banks operate under several kinds of risk at once. Fraudulent transactions cause direct financial loss, money laundering exposes institutions to regulatory and reputational damage, and customer churn erodes long-term revenue. Machine learning is widely used to flag all three, but the problems differ sharply in what a "positive" case looks like, how rare it is, and what a wrong prediction costs. A missed fraud or laundering case is expensive, while a false alarm consumes analyst time; in churn prediction, in contrast, a false alarm often only costs a retention offer.

A common difficulty is *class imbalance*. In the two transaction datasets studied here, fewer than 0.13 % of records are positive, so a classifier that labels everything as legitimate would be more than 99.8 % accurate while detecting nothing. Accuracy is therefore a poor yardstick, and the precision–recall trade-off is what matters in practice [1, 2].

Most published work treats fraud, anti-money-laundering (AML) and churn separately (see, e.g., [3, 4, 5]), which makes it hard to say whether conclusions about a model carry over from one risk to another. This study addresses that gap with a controlled comparison. The same three algorithms, the same imbalance-handling strategy, the same validation scheme and the same metrics are applied to three banking problems, so that differences in performance can be attributed mainly to the data and the risk rather than to the experimental setup.

Research question. *How effectively can machine-learning models detect different types of banking risk, and how does their performance vary across fraud detection, money laundering detection and customer churn prediction?*

The contributions of this paper are:

- a unified, leakage-aware pipeline applied to three banking risk datasets of very different size and imbalance;
- a controlled comparison of Decision Tree, Random Forest and XGBoost with SMOTE inside cross-validation, selected on PR-AUC and F1-score rather than accuracy;
- a cross-domain analysis of results, including training cost and the operational meaning of false positives and false negatives; and
- an explicit discussion of the limitations of the evaluation, including the near-perfect PaySim result and the CV-to-test shift on SAML-D.

II. Background and Related Work

A. Risk Detection in Banking

Fraud detection, AML monitoring and churn prediction are all supervised binary classification problems on tabular data, but they differ in structure. Fraud detection is a transaction-level problem in which signals such as inconsistent balance changes or unusual transaction types matter. AML monitoring is concerned with patterns of movement across accounts, and is typically evaluated at the transaction level when labels are available. Churn prediction operates at the customer level and relies on tenure, product holdings, balances and service history.

B. Class Imbalance and Resampling

Learning from imbalanced data is a well-studied problem [6]. The Synthetic Minority Over-sampling Technique (SMOTE) creates new minority examples by interpolating between a minority sample and its nearest minority neighbours [7]. Resampling must be applied only to training data; if it is applied before splitting or before cross-validation, synthetic points derived from validation records can leak into training and inflate scores. In this study SMOTE is therefore placed inside the cross-validation pipeline.

C. Tree-Based Classifiers

Decision trees [8] are interpretable but high-variance. Random Forests [9] average many decorrelated trees and are

robust on tabular data. Gradient-boosted trees, of which XGBoost [10] is a widely used implementation, build trees sequentially to correct earlier errors and often achieve the best accuracy on structured data. These three models span the range from an interpretable baseline to a strong ensemble.

D. Evaluation Under Imbalance

Because the precision–recall curve focuses on the positive class, it is generally more informative than the ROC curve on heavily imbalanced data [2, 1]. We therefore use the area under the precision–recall curve (PR-AUC), computed as average precision, as the primary selection metric, with F1-score as the tie-breaker, and report ROC-AUC, precision, recall and accuracy for completeness.

E. Datasets in the Literature

PaySim is a simulator of mobile-money transactions built from aggregated real logs [11]. SAML-D is a synthetic transaction-monitoring dataset created to support AML research, with labelled laundering typologies [12]. The Botswana bank customer churn dataset is publicly distributed through Kaggle [13]. Using these three together allows a like-for-like comparison across risk types.

F. Recent Studies (2024–2025)

Fraud. Mulyana and Ulum [3] compared Decision Tree, Random Forest and XGBoost on PaySim with random undersampling and reported that Random Forest was the most accurate (about 99%), followed by XGBoost (98.9%) and Decision Tree (97%). Doddamani *et al.* [14] oversampled PaySim and compared Logistic Regression, Gradient Boosting and XGBoost; XGBoost performed best (accuracy 99.88%), and a threshold-moving step was used to tune detection. Tetik and Gül [15] combined a Local Outlier Factor stage with LightGBM and SMOTE on PaySim. On a credit-card fraud dataset, Andrade-Arenas and Yactayo-Arias [16] found that Random Forest and XGBoost with SMOTE outperformed the other classifiers and reported AUPRC alongside accuracy and F1, noting that the age of the data limits generalization.

Money laundering. The SAML-D authors later proposed Tab-AML, a transformer-based model that reached an ROC-AUC of 93.01 with a 98% true-positive rate and a 51% false-positive rate on SAML-D, outperforming Tab-Transformer and XGBoost [4]. An interview study with eight AML specialists reported that rule-based transaction monitoring produces high false-positive rates and suggested graph analysis and anomaly detection as improvements [17].

Churn. Andespa *et al.* [5] compared Random Forest and XGBoost on a bank churn dataset with SMOTE and SMOTE-ENN, and found XGBoost with SMOTE to be best (F1 $= 0.873$, AUC $= 0.983$). Imani *et al.* [18] varied the churn rate from 15% to 1% in a telecom setting and found that XGBoost with SMOTE gave the highest F1-score at every imbalance level, while Random Forest performed poorly under severe imbalance.

Gap. Each of these studies addresses a single risk domain, and some rely on random splits [3, 16]. Our work instead applies one leakage-aware protocol, with time-ordered splits for the transaction data, to all three domains, which allows a direct cross-domain comparison.

Table I
DATASET OVERVIEW

	PaySim	SAML-D	Churn
Risk problem	Fraud	Laundering	Churn
Records	6,362,620	9,504,852	115,640
Positive cases	8,213	9,873	14,094
Positive rate	0.129%	0.104%	12.19%
Train records	4,453,834	6,653,396	80,948
Test records	1,908,786	2,851,456	34,692
Train positives	3,643	6,764	9,866
Test positives	4,570	3,109	4,228
Train / test rate	0.082 / 0.239%	0.102 / 0.109%	12.19 / 12.19%
Split type	Chronological	Chronological	Stratified

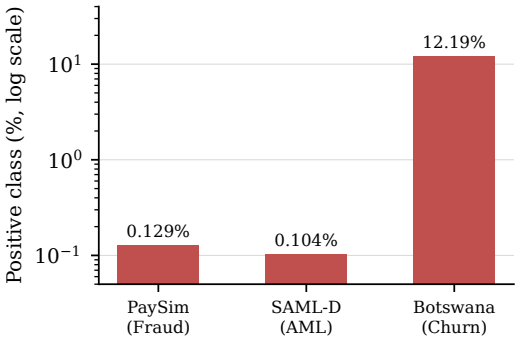


Fig. 1. Positive-class rate in each dataset (log scale). Fraud and laundering cases are roughly two orders of magnitude rarer than churn cases.

III. DATASETS

Table I summarizes the three datasets, and Fig. 1 compares their positive-class rates. PaySim and SAML-D are extremely imbalanced (0.129 % and 0.104 % positive), whereas churn is moderately imbalanced (12.19 %).

PaySim (fraud). Each record is a mobile-money transaction with a time step (hour), transaction type, amount, origin and destination balances before and after the transaction, and a fraud label. The original `isFlaggedFraud` field was removed, as it is a rule-based flag generated by the simulator rather than an input feature. No missing values or duplicates were found.

SAML-D (money laundering). Each record is a transaction with date, time, sender and receiver accounts, amount, payment and received currency, sender and receiver bank locations, payment type, a laundering label, and a laundering-typology label. The typology label was excluded, since it only exists for laundering cases and would leak the target.

Botswana churn. Each record is a customer with demographic attributes, income, tenure, credit information, outstanding loans, balance, number of products and complaints, and a churn flag. `Churn Reason` and `Churn Date` are populated only for churned customers (101,546 missing values, equal to the number of retained customers) and were removed to prevent target leakage.

IV. METHODOLOGY

Fig. 2 shows the pipeline applied to every dataset. The same sequence of steps is used throughout; only the fea-

ture engineering and a small number of model settings are dataset-specific.

A. Feature Engineering

Features were designed per dataset to capture the signals most relevant to each risk (Table II). Identifiers (account names, customer IDs, names, addresses and contact details) were removed.

In PaySim and SAML-D the running counts are computed with a cumulative count over the chronologically sorted data, so each record only sees earlier activity of the same account and no information flows backwards from the future.

B. Train–Test Split

The two transaction datasets are sorted by time and split chronologically, with the first 70% used for training and the last 30% for testing. This mimics deployment, where a model trained on the past scores future transactions. The churn dataset has no temporal ordering, so a stratified random 70/30 split (seed 42) was used, which preserves the 12.19% churn rate in both partitions.

C. Preprocessing and Imbalance Handling

Categorical variables are one-hot encoded (unseen categories ignored). For PaySim and churn, numerical variables are standardized; for SAML-D they are passed through unchanged. SMOTE is then applied to the *training portion of each fold only*, using an `imblearn` pipeline in which preprocessing, SMOTE and the classifier are fitted together, so validation folds are never resampled. The sampling ratio (minority-to-majority after resampling) was 0.1 for PaySim and SAML-D, which are extremely imbalanced, and 0.5 for churn. For PaySim, SMOTE used $k = 5$ neighbours.

D. Models

Three classifiers are compared in every domain: a Decision Tree (interpretable baseline), a Random Forest (bagged ensemble) and XGBoost (gradient boosting). Their hyperparameters (Table III) were fixed in advance and not tuned per dataset beyond adjusting tree depth and number of trees to dataset size. All use random seed 42.

E. Validation and Model Selection

On each training set, stratified five-fold cross-validation (shuffled, seed 42) was run for all three models with SMOTE inside each fold. The model with the highest mean PR-AUC was selected, with F1-score used as the secondary criterion. The selected model was then retrained on the entire training set (with SMOTE) and evaluated exactly once on the held-out test set.

F. Evaluation Metrics

With true positives TP, false positives FP, false negatives FN and true negatives TN:

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}, \tag{1}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{2}$$

Precision measures how many raised alerts are correct, and recall measures how many true cases are found. PR-AUC summarizes the precision–recall trade-off across all thresholds, and ROC-AUC summarizes the true-positive versus false-positive trade-off. All thresholds for hard predictions are the default 0.5 on predicted probability; no threshold tuning was performed.

V. Results

A. Cross-Validation Comparison

Table IV reports the five-fold cross-validation results for all three models, and Fig. 3 shows PR-AUC by model. The selected model in each domain is marked in bold.

On PaySim the top two models are very close: Random Forest has a mean PR-AUC of 0.9948 versus 0.9943 for XGBoost, a gap of 0.0005. Random Forest was selected because the protocol ranks on PR-AUC first, even though XGBoost had a higher CV F1-score (0.9930 vs. 0.9862) and higher precision, and ran about nine times faster (736 s vs. 6,929 s). On SAML-D XGBoost had the best PR-AUC, ROC-AUC and F1, while Random Forest had the highest precision and the Decision Tree the highest recall. On churn XGBoost led on every metric, while Random Forest was clearly the weakest ($F_1 = 0.827$).

B. Final Test Performance

Table V gives the held-out test results for the selected models, and Fig. 4 compares precision, recall and F1.

PaySim and churn are detected almost perfectly. SAML-D is clearly harder: recall is high (92.70 %), but precision is only 74.13 %, so about one in four alerts is a false alarm. Note that SAML-D accuracy is 99.96 %, higher than churn accuracy (99.44 %), even though SAML-D is by far the weaker detector. This is a direct illustration of why accuracy is misleading under severe imbalance.

C. Error Analysis

Fig. 5 shows the confusion matrices. On PaySim only one legitimate transaction was flagged and one fraud missed among 1,908,786 test records. On SAML-D, 2,882 of 3,109 laundering transactions were detected, 227 were missed and 1,006 legitimate transactions were flagged. On churn, 4,163 of 4,228 churners were identified, with 65 missed and 130 false alarms.

D. Cross-Validation Versus Test

Fig. 6 compares cross-validated and test scores. Churn is stable (PR-AUC 0.9982 $\rightarrow$ 0.9986; F1 0.9751 $\rightarrow$ 0.9771). PaySim improves on the test set (PR-AUC 0.9948 $\rightarrow$ 0.9999). SAML-D shows the largest shift: PR-AUC rises from 0.827 to 0.936, while F1 slightly falls (0.840 $\rightarrow$ 0.824) and the balance of precision and recall changes completely (CV precision 0.982 and recall 0.734; test precision 0.741 and recall 0.927). This pattern is discussed in Section VI.

E. Computational Cost

Fig. 7 shows the training and testing time of the final models. PaySim's Random Forest took about 30 minutes to train on 4.45M rows, SAML-D's XGBoost about 10 minutes on 6.65M rows, and churn's XGBoost under 45 seconds. SAML-D had the longest scoring time (43 s for 2.85M rows). Computational cost therefore tracks dataset size and model family and is relevant to deployment choices.

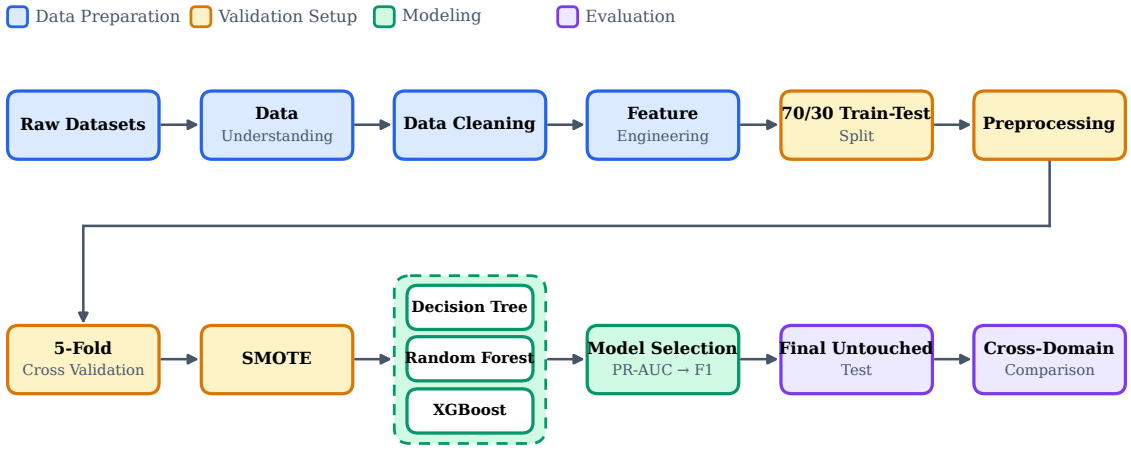


Fig. 2. Overview of the methodology pipeline shared by all three banking problems, from raw data through preprocessing, SMOTE-based cross-validation, model selection and the final untouched test.

Table II
ENGINEERED FEATURES AND LEAKAGE CONTROLS PER DATASET

Dataset	Engineered features	Removed (leakage / identifiers)
PaySim	Origin and destination balance errors (old − amount − new and old + amount − new); zero-balance flags; log transforms of amount and balances; amount-to-balance ratios; running transaction counts for sender and receiver; same-account flag; transfer/cash-out flag. (1 categorical + 21 numerical features.)	Sender and receiver names, `isFlaggedFraud`.
SAML-D	Currency mismatch; bank-location mismatch and same-location flag; log amount and zero-amount flag; hour, day of week and weekend flag; running counts for sender, receiver and sender–receiver pair. (5 categorical + 12 numerical features.)	`Laundering_type`, sender and receiver accounts, raw date and time.
Churn	Age (from date of birth); balance-to-income and loan-to-income ratios; log of balance, income and loans; products per tenure; balance per product; complaints per tenure. (6 categorical + 18 numerical features.)	`Churn Reason`, `Churn Date`, IDs, names, address, contact information, date of birth.

Table III
MODEL HYPERPARAMETERS

Model	Setting	PaySim	SAML-D	Churn
Decision Tree	max depth	12	none	10
	min leaf samples	10	2	10
Random Forest	trees	100	200	200
	max depth	20	none	20
	min leaf samples	5	2	5
XGBoost	trees	200	300	300
	max depth	6	6	6
	learning rate	0.10	0.10	0.05
	subsample / colsample	0.8 / 0.8	0.8 / 0.8	0.8 / 0.8
SMOTE	sampling ratio	0.1	0.1	0.5

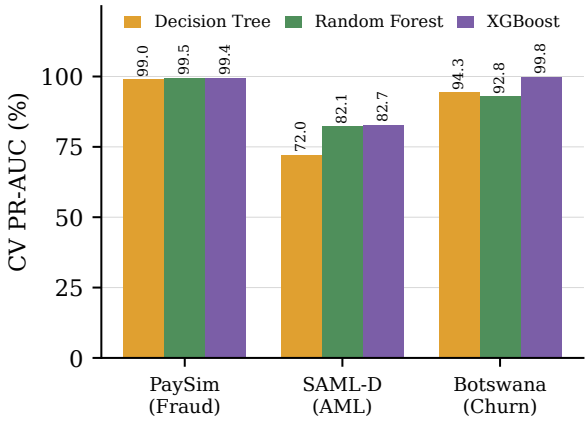


Fig. 3. Cross-validated PR-AUC of the three classifiers in each domain. XGBoost leads on SAML-D and churn; on PaySim, Random Forest and XGBoost are almost tied.

VI. DISCUSSION

No single best model. Random Forest was selected for fraud, and XGBoost for money laundering and churn. The PaySim margin was very small, so the stronger claim is not that Random Forest is better for fraud but that the ranking of models was not the same in all three domains and that the size of the gap between them varies widely (large on churn, small on PaySim). This supports the central message that model choice should be made per problem.

Relation to recent studies. Our churn results agree with recent work in which XGBoost with SMOTE was the strongest model and Random Forest degraded under imbalance [5, 18]: in our CV, XGBoost led on churn while Random Forest was weakest. The PaySim ordering (Random Forest $\geq$ XGBoost $>$ Decision Tree on PR-AUC) matches the ordering reported in [3], although their protocol (undersampling, random split) differs from ours. The Tab-AML result on SAML-D [4] is not directly comparable to ours, because the split, features and operating point differ; it does, however, point to the same difficulty we observe, namely a

Table IV

FIVE-FOLD CROSS-VALIDATION RESULTS (SMOTE INSIDE FOLDS). MEAN OVER FOLDS; SELECTED MODEL IN BOLD.

Dataset	Model	Acc.	Prec.	Rec.	F1	ROC-AUC	PR-AUC	CV time (s)
PaySim	Decision Tree	0.99990	0.8952	0.9920	0.9407	0.9980	0.9897	571
	Random Forest	**0.99998**	**0.9810**	**0.9915**	**0.9862**	**0.9991**	**0.9948**	**6,929**
	XGBoost	0.99999	0.9950	0.9909	0.9930	0.9992	0.9943	736
SAML-D	Decision Tree	0.99968	0.8965	0.7725	0.8298	0.8925	0.7205	448
	Random Forest	0.99965	0.9885	0.6634	0.7938	0.9939	0.8212	5,840
	XGBoost	**0.99972**	**0.9824**	**0.7343**	**0.8404**	**0.9955**	**0.8267**	**2,464**
Churn	Decision Tree	0.9804	0.9033	0.9400	0.9212	0.9841	0.9428	39
	Random Forest	0.9545	0.7708	0.8923	0.8270	0.9863	0.9283	117
	XGBoost	**0.9939**	**0.9682**	**0.9822**	**0.9751**	**0.9997**	**0.9982**	**155**

Table V

FINAL HELD-OUT TEST PERFORMANCE OF THE SELECTED MODEL IN EACH DOMAIN (%)

Dataset	Model	Acc.	Prec.	Rec.	F1	ROC-AUC	PR-AUC	Train (s)	Test (s)
PaySim (Fraud)	Random Forest	99.9999	99.9781	99.9781	99.9781	100.0000	99.9999	1,812.9	19.6
SAML-D (AML)	XGBoost	99.9568	74.1255	92.6986	82.3782	99.9475	93.5894	592.6	43.1
Churn	XGBoost	99.4379	96.9718	98.4626	97.7115	99.9803	99.8634	43.0	0.6

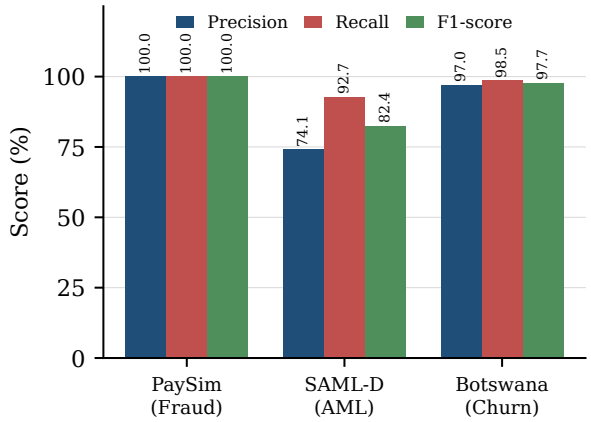


Fig. 4. Test precision, recall and F1-score of the selected model in each domain. Precision is the weak point on SAML-D.

trade-off between detecting laundering and generating false alerts.

Why the three problems behave differently. PaySim's engineered balance-error features expose a strong, almost deterministic signature of fraud in the simulated data, which likely explains the near-perfect test scores; whether real-world fraud is as separable is not something this experiment can show. Churn benefits from informative customer-behaviour features and a moderate 12.19% positive rate. SAML-D is the hardest: laundering is defined by patterns across accounts and time, while the features used here are mostly transaction-level, so many laundering transactions look ordinary in isolation. Richer network or time-window features are a natural next step.

Operational meaning of the errors. On SAML-D, 1,006 false alerts accompany 2,882 true detections. For an AML team this is roughly 0.35 false alert per correct one, which may be acceptable if missing laundering is much costlier than reviewing an alert, but it must be a deliberate choice. A decision threshold other than 0.5 could trade recall for precision; this was not explored. On churn, the cost of a false positive (an unnecessary retention offer) is low, so the high recall is appropriate.

The SAML-D shift from CV to test. In cross-validation SAML-D had high precision (0.98) and modest recall (0.73); on the test set the pattern reversed. We have not isolated the cause. Two plausible contributors are that cross-validation shuffles chronologically ordered training data while the test set is a later time period, and that the laundering mix may differ between periods. This is a reason to treat the SAML-D CV numbers as a guide to model ranking rather than as an estimate of test performance.

Accuracy is not informative here. Accuracy exceeded 99.4% in all three domains, but F1 ranged from 82.4% to 99.98%. PR-AUC, precision and recall expose differences that accuracy hides, supporting the use of PR-AUC for selection.

VII. LIMITATIONS

1. **Near-perfect PaySim performance.** With one false positive and one false negative, and ROC-AUC of 1.0, the PaySim result should be read with care. The data are simulated, the test fraud rate (0.239%) differs from the training rate (0.082%), and the result does not establish that real fraud is this easy to detect.
2. **Synthetic or simulated data.** PaySim and SAML-D are simulation-based, so performance may not transfer to production systems.
3. **Fixed hyperparameters.** Hyperparameters were set once per dataset rather than tuned with a search, so the ranking of models could change under tuning.
4. **Single split and seed.** Results come from one train–test split and one random seed, and no confidence intervals or significance tests are reported.
5. **Default decision threshold.** No threshold tuning or cost-sensitive analysis was done, although costs of false positives and negatives differ by domain.
6. **Shuffled CV on chronological data.** For PaySim and SAML-D, cross-validation inside the training period was

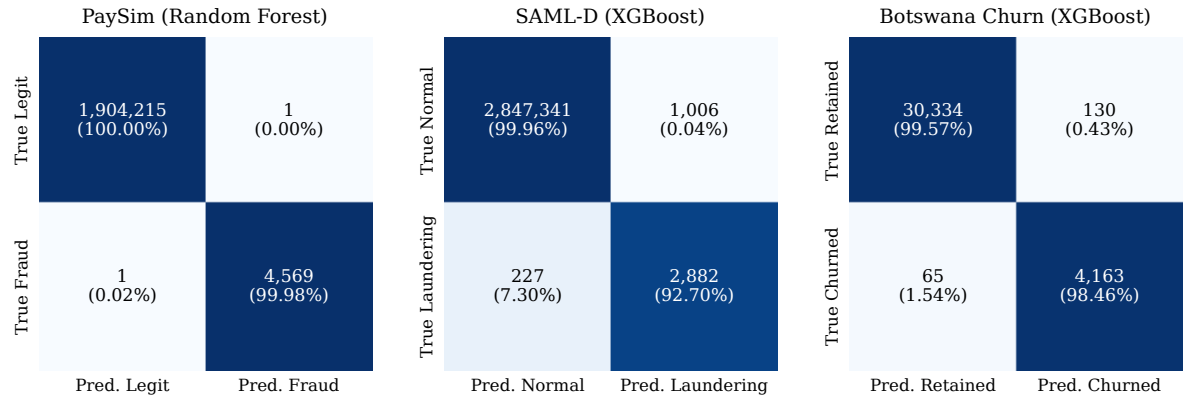


Fig. 5. Confusion matrices on the held-out test sets (counts and row-normalized percentages).

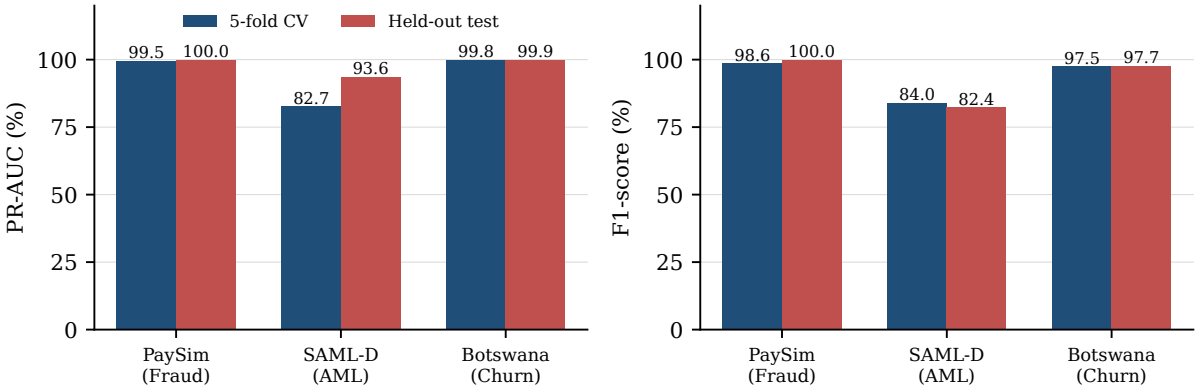


Fig. 6. Cross-validation versus held-out test PR-AUC (left) and F1-score (right).

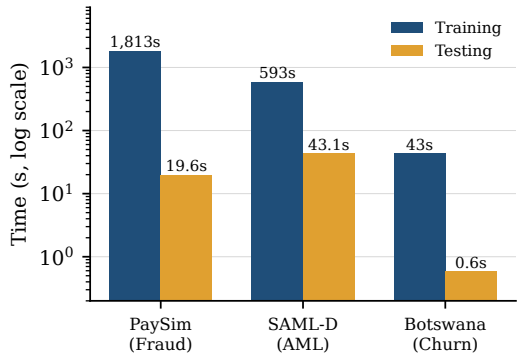


Fig. 7. Training and testing time of the selected final models (log scale).

shuffled rather than time-ordered, which may explain part of the CV-to-test differences.

7. **Transaction-level features for AML.** SAML-D features do not include graph or network features, which are central to laundering detection.
8. **No interpretability analysis.** Feature importance and explanations were not examined.

VIII. Future Work

Future work should (i) tune hyperparameters with a proper search and report variance across seeds; (ii) use time-aware cross-validation for the transaction datasets; (iii) add graph-based and time-window features for AML; (iv) tune decision thresholds against explicit cost assumptions; (v) add interpretability analysis such as feature importance; (vi) compare SMOTE with class weighting and other resampling strategies; and (vii) validate on real or additional datasets.

IX. Conclusion

This paper compared Decision Tree, Random Forest and XGBoost for fraud, money laundering and churn detection under one leakage-aware framework with SMOTE inside cross-validation and PR-AUC-based model selection. Random Forest was selected for PaySim and XGBoost for SAML-D and churn. Fraud and churn were detected with F1-scores of 99.98 % and 97.71 %, while money laundering remained harder (F1 82.38 %, precision 74.13 %, recall 92.70 %). Accuracy was above 99.4 % in all cases and so could not distinguish the problems. The main conclusion is that performance and the best model are problem-specific: no model was universally best, and the right choice depends on the data, the risk, and the balance between missed cases and false alarms. The results should be interpreted within the stated limitations, in particular the simulated nature of two datasets and the near-perfect PaySim score.

References

[1] J. Davis and M. Goadrich, “The relationship between precision-recall and ROC curves,” in *Proc. 23rd Int. Conf. Machine Learning (ICML)*, 2006, pp. 233–240.

[2] T. Saito and M. Rehmsmeier, “The precision-recall plot is

more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, e0118432, 2015.

[3] I. Mulyana and M. B. Ulum, "Application of machine learning models for fraud detection in synthetic mobile financial transactions," *J. Ilmu Pengetahuan dan Teknol. Komput. (JITK)*, vol. 10, no. 4, pp. 759–768, May 2025.

[4] B. Oztas, D. Cetinkaya, F. Adedoyin, M. Budka, H. Dogan, and G. Aksu, "Tab-AML: A transformer based transaction monitoring model for anti-money laundering," in *Proc. IEEE Conf. Artificial Intelligence (CAI)*, Santa Clara, CA, USA, 2025, pp. 161–167.

[5] R. Andespa, K. Sadik, C. Suhaeni, and A. M. Soleh, "Evaluating random forest and XGBoost for bank customer churn prediction on imbalanced data using SMOTE and SMOTE-ENN," *Media Statistika*, vol. 18, no. 1, pp. 25–36, Oct. 2025.

[6] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, 2009.

[7] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.

[8] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, *Classification and Regression Trees*. Belmont, CA, USA: Wadsworth, 1984.

[9] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.

[10] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[11] E. A. Lopez-Rojas, A. Elmir, and S. Axelsson, "PaySim: A financial mobile money simulator for fraud detection," in *Proc. 28th European Modeling and Simulation Symp. (EMSS)*, Larnaca, Cyprus, 2016.

[12] B. Oztas, D. Cetinkaya, F. Adedoyin, M. Budka, H. Dogan, and G. Aksu, "Enhancing anti-money laundering: Development of a synthetic transaction monitoring dataset," in *Proc. IEEE Int. Conf. e-Business Engineering (ICEBE)*, 2023, pp. 47–54.

[13] S. D. Mfazi, "Bank customer churn," Kaggle dataset.

[14] S. S. Doddamani, K. K. Girish, and B. Bhowmik, "Money laundering detection in imbalanced e-wallet transactions with threshold optimization," in *Proc. 9th IEEE Int. Conf. Convergence in Technology (I2CT)*, 2024.

[15] A. Tetik and E. Gül, "Fraud detection using machine learning in banking operations," in *Proc. Innovations in Intelligent Systems and Applications Conf. (ASYU)*, Bursa, Türkiye, 2025.

[16] L. Andrade-Arenas and C. Yactayo-Arias, "Comparative analysis of machine learning models for credit card fraud detection using SMOTE for class imbalance," *Int. J. Saf. Secur. Eng.*, vol. 15, no. 5, pp. 893–901, May 2025.

[17] B. Oztas, D. Cetinkaya, F. Adedoyin, M. Budka, G. Aksu, and H. Dogan, "Transaction monitoring in anti-money laundering: A qualitative analysis and points of view from industry," *Future Gener. Comput. Syst.*, vol. 159, pp. 161–171, 2024.

[18] M. Imani, A. Beikmohammadi, and H. R. Arabnia, "Comprehensive analysis of random forest and XGBoost performance with SMOTE, ADASYN, and GNUS under varying imbalance levels," *Technologies*, vol. 13, no. 3, art. 88, Feb. 2025.